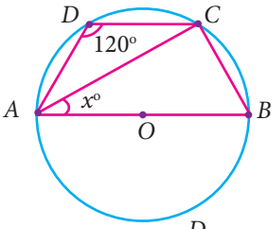
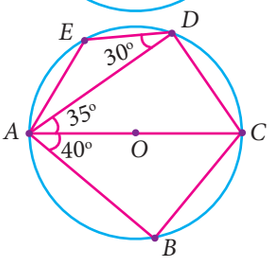
Exercise 4.4

1. Find the value of x in the given figure.

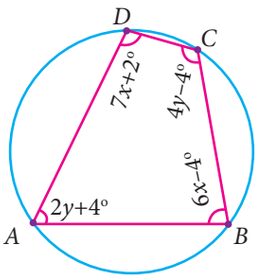


2. In the given figure, AC is the diameter of the circle with centre O. If $\angle ADE = 30°$; $\angle DAC = 35°$ and $\angle CAB = 40°$.

   Find (i) $\angle ACD$ (ii) $\angle ACB$ (iii) $\angle DAE$

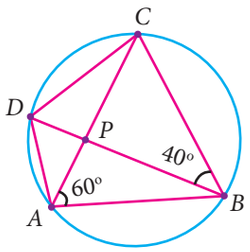


3. Find all the angles of the given cyclic quadrilateral $ABCD$ in the figure.

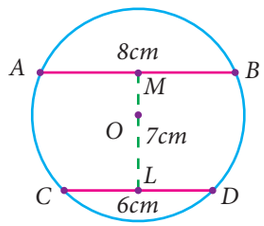


4. In the given figure, $ABCD$ is a cyclic quadrilateral where diagonals intersect at P such that $\angle DBC = 40°$ and $\angle BAC = 60°$ find

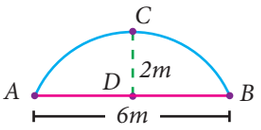
   (i) $\angle CAD$ (ii) $\angle BCD$



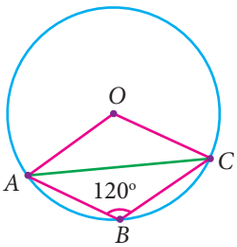
5. In the given figure, AB and CD are the parallel chords of a circle with centre O. Such that AB = 8cm and CD = 6cm. If $OM \perp AB$ and $OL \perp CD$ distance between LM is 7cm. Find the radius of the circle?



6. The arch of a bridge has dimensions as shown, where the arch measure 2m at its highest point and its width is 6m. What is the radius of the circle that contains the arch?



7. In figure, $\angle ABC = 120^0$, where A, B and C are points on the circle with centre O. Find $\angle OAC$?



8. A school wants to conduct tree plantation programme. For this a teacher allotted a circle of radius 6m ground to nineth

4-Geometry_Term1.indd 173 02/12/2022 21:42:47